

Cheat Sheet $\mathbb{F}_9$

June 23, 2026

Extension field generated by the polynomial

$$X^2 + X + 2$$

The numerical value of the polynomial is 14

The field is equianharmonic, $a = 2$

The polynomial used to define the field is : $X^2 + X + 2 = 14$

$$Z_i = \log_\alpha(1 + \alpha^i)$$

Subfields in Detail

The subfield $\mathbb{F}_3$, divisor $h = 1$, is generated by polynomial 4:

Subfield of Order 3:

polynomial:

Field basis:

$$(1, 7)$$

i	elt in Fq	elt in FQ
0	0	0
1	1	1
2	2	2

The power table shows $(\alpha^d)^i$ for $i = 0, 1, \dots, e$ where $q = p^e$: $q = 9$, $p = 3$, $e = 2$, $d = 4$, $\alpha = 3$,

i	$i \cdot d$	α^{id}	vector
0	0	1	(0, 1)
1	4	2	(0, 2)

Table of Elements of $\mathbb{F}_9$

0	00	0
1	01	1
2	02	2
3	10	α
4	11	$\alpha + 1$
5	12	$\alpha + 2$
6	20	2α
7	21	$2\alpha + 1$
8	22	$2\alpha + 2$

i	γ_i	$-\gamma_i$	γ_i^{-1}	$\log_\alpha(\gamma_i)$	α^i	Z_i	$\phi(\gamma_i)$	$T(\gamma_i)$	$N(\gamma_i)$
0	$0 = 0$	0	DNE	DNE	1	4	0	0	0
1	$1 = 1$	2	1	0	3	7	1	2	1
2	$2 = \alpha^4$	1	2	4	7	3	2	1	1
3	$\alpha = \alpha$	6	4	1	8	5	8	2	2
4	$\alpha + 1 = \alpha^7$	8	3	7	2	DNE	6	1	2
5	$\alpha + 2 = \alpha^6$	7	7	6	6	2	7	0	1
6	$2\alpha = \alpha^5$	3	8	5	5	1	4	1	2
7	$2\alpha + 1 = \alpha^2$	5	5	2	4	6	5	0	1
8	$2\alpha + 2 = \alpha^3$	4	6	3	1	4	3	2	2

+	0	1	2	3	4	5	6	7	8
0	0	1	2	3	4	5	6	7	8
1	1	2	0	4	5	3	7	8	6
2	2	0	1	5	3	4	8	6	7
3	3	4	5	6	7	8	0	1	2
4	4	5	3	7	8	6	1	2	0
5	5	3	4	8	6	7	2	0	1
6	6	7	8	0	1	2	3	4	5
7	7	8	6	1	2	0	4	5	3
8	8	6	7	2	0	1	5	3	4

$\cdot$	1	2	3	4	5	6	7	8
1	1	2	3	4	5	6	7	8
2	2	1	6	8	7	3	5	4
3	3	6	7	1	4	5	8	2
4	4	8	1	5	6	2	3	7
5	5	7	4	6	2	8	1	3
6	6	3	5	2	8	7	4	1
7	7	5	8	3	1	4	2	6
8	8	4	2	7	3	1	6	5

Cyclic structure

i	α^i	α^i	vector	reduced rep.
0	3^0	1	01	1
1	3^1	3	10	α
2	3^2	7	21	$2\alpha + 1$
3	3^3	8	22	$2\alpha + 2$
4	3^4	2	02	2
5	3^5	6	20	2α
6	3^6	5	12	$\alpha + 2$
7	3^7	4	11	$\alpha + 1$
8	3^8	1	01	1

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0	3^0	1	01
1	3^1	3	10
2	3^2	7	21
3	3^3	8	22
4	3^4	2	02
5	3^5	6	20
6	3^6	5	12
7	3^7	4	11
8	3^8	1	01

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1	$1 = 1$	2	1	0	3	7	1	2	1
2	$2 = \alpha^4$	1	2	4	7	3	2	1	1
3	$\alpha = \alpha$	6	4	1	8	5	8	2	2
4	$\alpha + 1 = \alpha^7$	8	3	7	2	DNE	6	1	2
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8	$2\alpha + 2 = \alpha^3$	4	6	3	1	4	3	2	2

+	0	1	2	3	4	5	6	7	8
0	0	1	2	3	4	5	6	7	8
1	1	2	0	4	5	3	7	8	6
2	2	0	1	5	3	4	8	6	7
3	3	4	5	6	7	8	0	1	2
4	4	5	3	7	8	6	1	2	0
5	5	3	4	8	6	7	2	0	1
6	6	7	8	0	1	2	3	4	5
7	7	8	6	1	2	0	4	5	3
8	8	6	7	2	0	1	5	3	4

·	1	2	3	4	5	6	7	8
1	1	2	3	4	5	6	7	8
2	2	1	6	8	7	3	5	4
3	3	6	7	1	4	5	8	2
4	4	8	1	5	6	2	3	7
5	5	7	4	6	2	8	1	3
6	6	3	5	2	8	7	4	1
7	7	5	8	3	1	4	2	6
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3	3^3	8	22
4	3^4	2	02
5	3^5	6	20
6	3^6	5	12
7	3^7	4	11
8	3^8	1	01